

Chapter 40: Preeclampsia Syndrome — Revision Notes

Williams Obstetrics 26e, pp. 1777–1839 of the PDF. High-yield numbers are in **bold**. The four tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 14 figures are included from the PDF.

Big picture: Hypertensive disorders complicate **up to 10%** of pregnancies and are one of the **deadly triad** (with hemorrhage and infection). Preeclampsia, alone or superimposed on chronic hypertension, is the most dangerous: **7%** of US pregnancy-related deaths (2011–2015), and most are **preventable**. It is a **two-stage disorder**: faulty trophoblastic remodeling of spiral arterioles (placental syndrome) → systemic **endothelial cell activation** (maternal syndrome) → vasospasm, capillary leak, and ischemic/thrombotic organ damage.

1. Terminology and Diagnosis

ACOG classification: four types of hypertensive disease

1. **Preeclampsia and eclampsia syndrome**
2. **Chronic hypertension** of any etiology
3. **Preeclampsia superimposed on chronic hypertension**
4. **Gestational hypertension**: no definitive preeclampsia, and BP **resolves by 12 wk postpartum**

Diagnosing hypertension

- **Systolic ≥ 140 and/or diastolic ≥ 90 mm Hg**; diastolic = **Korotkoff phase V**.
- The old criteria (**rise of 30 systolic / 15 diastolic** above midpregnancy values) are **no longer used**, but surveillance is reasonable: some women have eclampsia with BP **$<140/90$** .
- **"Delta hypertension"** = a relatively acute rise in BP that still lies in the normal range. It may signal preeclampsia; some of these women develop eclampsia or HELLP.
- 140/90 dates from the 1950s and was chosen by **insurance companies for middle-aged men**; pregnancy-specific normal ranges are being developed.

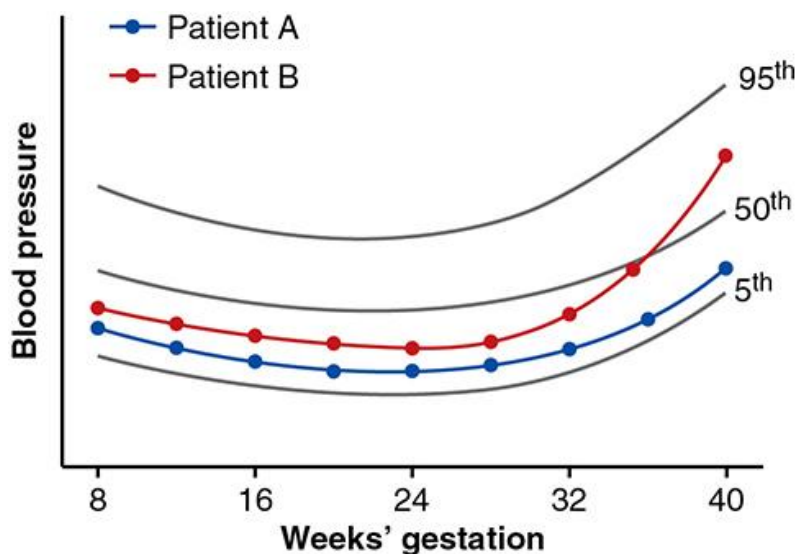


Figure 40-1. "Delta hypertension": patient B rises from the 25th to the 75th percentile after ~36 wk yet is still "normotensive."

Gestational hypertension

- BP **$\geq 140/90$ for the first time after midpregnancy, without proteinuria**.
- **Almost half** go on to develop preeclampsia.

- ⚠️ **10% of eclamptic seizures occur before overt proteinuria** is detected → do not ignore a rising BP just because protein is absent.
- Called **transient hypertension** if preeclampsia never develops and BP normalizes by **12 wk postpartum**.

Preeclampsia syndrome

- A **pregnancy-specific syndrome that can affect virtually every organ system**.
- Proteinuria remains a primary criterion: an objective marker of the **system-wide endothelial leak**. But some women have neither proteinuria nor FGR, so other criteria are accepted (Table 40-1).

Onset category	Gestational age
Early onset	<34 wk
Late onset	≥34 wk
Preterm onset	<37 wk
Term onset	≥37 wk

Table 40-1. Classification and Diagnosis of Pregnancy-Associated Hypertension

Condition	Criteria Required
Gestational hypertension	BP >140/90 mm Hg after 20 weeks in previously normotensive women
Preeclampsia: Hypertension plus	
Proteinuria	≥300 mg/24 h, or urine protein:creatinine ratio ≥0.3, or dipstick 1+ persistent ^a
	or
Thrombocytopenia	Platelets <100,000/μL
Renal insufficiency	Creatinine >1.1 mg/dL or doubling of baseline ^b
Liver involvement	Serum transaminase levels ^c twice normal
Cerebral symptoms	Headache, visual disturbances, convulsions
Pulmonary edema	—

^aRecommended only if sole available test. ^bNo prior renal disease. ^cAspartate transaminase (AST) or alanine transaminase (ALT). BP = blood pressure.

The table prints ">140/90"; the text defines hypertension as 140/90 "or greater", which is the standard ACOG threshold.

Severity

- The Task Force **discourages "mild preeclampsia"**; there is no consensus "moderate" category. Parkland uses **"severe" vs "nonsevere"** (ACOG 2020).
- **Ominous symptoms:**
 - **Headache or visual disturbance** → may precede eclampsia.
 - **Eclampsia** = a convulsion in a woman with preeclampsia not attributable to another cause. Generalized; before, during or after labor. **~10% occur >48 h postpartum**.
 - **Epigastric/RUQ pain** → hepatocellular necrosis, ischemia and edema (↑ transaminases).
 - **Thrombocytopenia** → worsening disease (platelet activation/aggregation, microangiopathic hemolysis).
- Profound signs usually cannot be temporized → delivery. ⚠️ Apparently mild disease can **progress rapidly** to severe.

Table 40-2. Indicators of Severity of Gestational Hypertensive Disorders^a

Abnormality	Nonsevere ^b	Severe
Diastolic BP	<110 mm Hg	≥110 mm Hg
Systolic BP	<160 mm Hg	≥160 mm Hg
Proteinuria ^c	None to positive	None to positive
Headache	Absent	Present
Visual disturbances	Absent	Present
Upper abdominal pain	Absent	Present
Oliguria	Absent	Present
Convulsion (eclampsia)	Absent	Present
Serum creatinine	Normal	Elevated
Thrombocytopenia (<100,000/μL)	Absent	Present
Serum transaminase elevation	Minimal	Marked
Fetal-growth restriction	Absent	Present
Pulmonary edema	Absent	Present
Gestational age	Late	Early

^aCompare with criteria in Table 40-1. ^bIncludes "mild" and "moderate" hypertension not specifically defined. ^cMost disregard degrees of proteinuria to classify as nonsevere or severe. BP = blood pressure.

Memory hook: severe = "**160/110**, plus any symptom or organ": head, eyes, belly, kidneys (oliguria/creatinine), platelets, liver, lungs, fetus (FGR). **Proteinuria amount does not count.**

Preeclampsia superimposed on chronic hypertension

- **Chronic hypertension** = BP ≥140/90 before pregnancy or before 20 wk, or both.
- Hard to classify when first seen after midpregnancy: BP often falls into the normal range before 20 wk and returns to hypertensive levels in the 3rd trimester; many have no end-organ evidence.
- **20–50%** of women with chronic hypertension have a clear BP rise, typically **after 24 wk**.
- **Diagnosis:** new-onset or worsening hypertension **plus new proteinuria or other Table 40-1 findings**.
- Compared with "pure" preeclampsia: **earlier onset, more severe, more often FGR**.

2. Incidence and Risk Factors

- Preeclampsia in **5–8%** of all pregnancies.
 - **Nulliparas 3–10%; multiparas 2–5%.**
 - Young and nulliparous women are most vulnerable; older women more often have **chronic hypertension with superimposed preeclampsia**.
- MFMU nulliparas: **5% white, 9% Hispanic, 11% African American**. Black women also have more severe adverse outcomes.
- Other risks: prior preeclampsia (**especially HELLP**), metabolic syndrome, hyperhomocysteinemia, chronic kidney disease.
- Lesser risks: **HIV, sleep-disordered breathing, male fetus**, affected family members, maternal and fetal genetics, **mirror syndrome**.
- ⚠️ **Smoking paradoxically lowers** the risk of hypertension in pregnancy.

Table 40-3. Selected Clinical Risk Factors for Preeclampsia

Risk Factor	Pooled Unadjusted Relative Risk ^a
Nulliparity	2.1 (1.9–2.4)
Age >35	1.2 (1.1–1.3)
BMI >30	2.8 (2.6–3.1)
Multifetal gestation	2.9 (2.6–3.1)
Prior abruption	2.0 (1.4–2.7)
Diabetes	3.7 (3.1–4.3)
Prior preeclampsia	8.4 (7.1–9.9)
CHTN	5.1 (4.0–6.5)
Prior stillbirth	2.4 (1.7–3.4)
Chronic kidney disease	1.8 (1.5–2.1)
ART	1.8 (1.6–2.1)
SLE	2.5 (1.0–6.3)
APA	2.8 (1.8–4.3)

^a95% confidence interval. APA = antiphospholipid antibody; ART = assisted reproductive technology; BMI = body mass index; CHTN = chronic hypertension; SLE = systemic lupus erythematosus. Data from >25 million pregnancies (Bartsch, 2016).

Eclampsia incidence

- In well-resourced countries **1 in 2000–3000 deliveries**; falls where care is accessible.
- Parkland: **~1 in 2000 births** now, attributed to prenatal care access and active management.

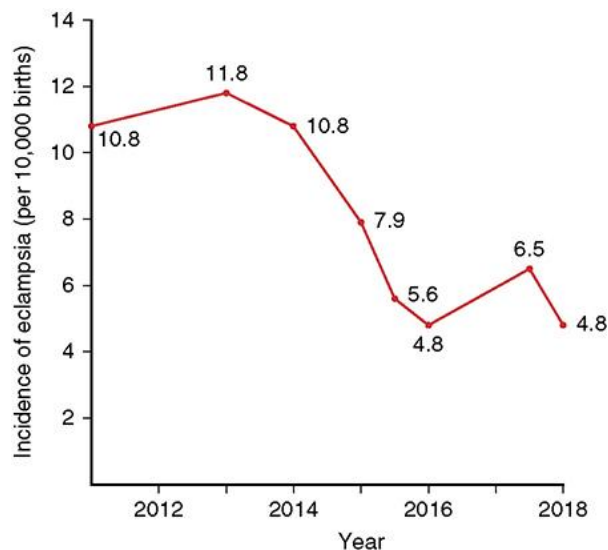


Figure 40-2. Eclampsia at Parkland fell from 10.8–11.8 to ~4.8 per 10,000 births (2011–2018).

3. Etiopathogenesis

Who gets it? Any theory must explain higher risk with:

- **First exposure to chorionic villi** (nulliparity)
- A **superabundance of villi**: twins, hydatidiform mole
- Preexisting **endothelial activation or inflammation**

- **Genetic predisposition**
- ⚠️ **A fetus is not required**; villi are essential but need not be intrauterine (e.g., advanced abdominal pregnancy).
- Final common pathway: **endothelial damage** → **vasospasm** → **plasma transudation** → **ischemic and thrombotic sequelae**.

Two-stage disorder (Redman)

- **Stage I (placental syndrome)**: faulty endovascular trophoblastic remodeling of spiral arterioles.
- **Stage II (maternal syndrome)**: systemic inflammatory response and endothelial activation. It is **modified by maternal conditions** with endothelial activation: chronic hypertension, renal disease, obesity, immunological/connective tissue disorders, diabetes.
- Staging is artificial: a spectrum, likely with "isoforms" (early vs late onset, different placental and genetic findings).

Proposed primary mechanisms

- Abnormal trophoblastic invasion of uterine vessels
- Dysfunctional **immunological tolerance** between maternal, paternal (placental) and fetal tissues
- Maternal **maladaptation** to the cardiovascular or inflammatory changes of pregnancy
- **Genetic** factors and epigenetic influences

Stage I: placental syndrome

- **Normal**: endovascular trophoblasts **replace the endothelial and muscular lining** of spiral arterioles in the decidua basalis → wide, low-resistance vessels. Veins are invaded only superficially.
- **Preeclampsia (some cases)**: invasion reaches **decidual but not myometrial** vessels. Myometrial arterioles keep their endothelium and musculoelastic tissue → **mean external diameter only half** of normal. More prevalent in **early-onset** disease. Soluble antiangiogenic factors likely play a role.
- Early placental changes: endothelial damage, insudation of plasma into vessel walls, myointimal proliferation, medial necrosis.
- **Atherosclerosis** (Hertig) = **lipid-laden myointimal cells and macrophages**. More common with preeclampsia **<34 wk**; may mark women at risk of later **cardiovascular disease**.
- Narrow lumens → reduced perfusion, **hypoxia** → systemic inflammatory response (stage II).
- Defective placentation also predisposes to gestational hypertension, **preterm delivery, FGR and abruption**.

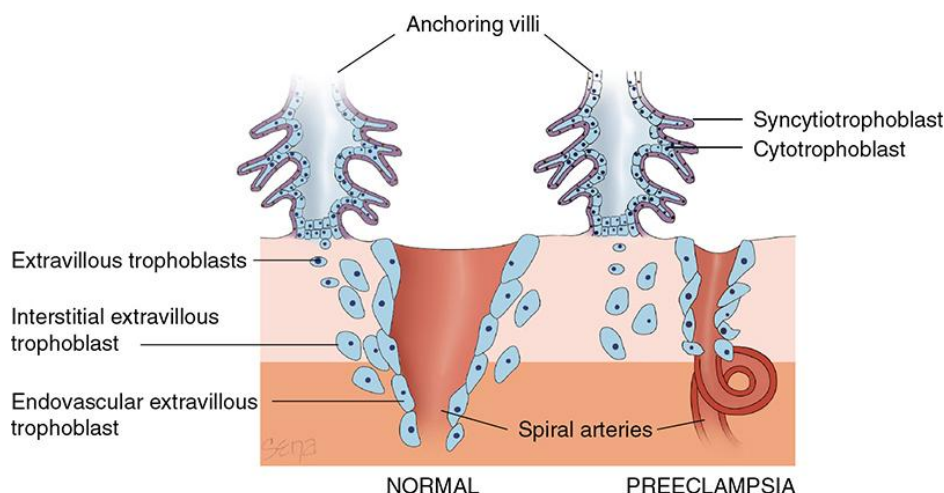


Figure 40-3. Normal: endovascular extravillous trophoblast converts spiral arteries into wide, low-resistance vessels. Preeclampsia: shallow invasion leaves narrow, high-resistance arteries.

Immunological factors

- Loss of maternal tolerance to paternal antigens; placental histology resembles **acute graft rejection**.
- **More paternal antigen** → **more risk**:
 - **Complete mole** (all-paternal diploid) → early-onset preeclampsia with later-stage moles.

- **Trisomy 13: 30–40% incidence**; the **sFlt-1 gene is on chromosome 13**.
- Prior pregnancy with the **same partner** may be "immunizing"; a **new partner raises risk**.
- Reduced **HLA-G** (immunosuppressive) on extravillous trophoblast early in affected pregnancies. The **1597ΔC** allele (more common in Black women) causes incomplete HLA-G expression.
- Normal pregnancy has a **Th2 bias**; in women who develop preeclampsia, **Th1 activity rises from the early 2nd trimester**.

Genetic factors

- **Multifactorial, polygenic**; hundreds of maternal and paternal genes; fetal genes also contribute. No single gene will explain it.

Relative of affected woman	Incident risk of preeclampsia
Daughters	20–40%
Sisters	11–37%
Twins	22–47%

- Latina women have a lower incidence (interaction of American Indian and white genes).

Stage II: maternal syndrome

Endothelial cell activation (centerpiece of pathogenesis)

- Placental antiangiogenic and metabolic factors and activated leukocytes → cytokines (**TNF-α, interleukins**) → **oxidative stress** (toxic oxygen radicals).
- Consequences:
 - ↓ **nitric oxide**, disturbed prostaglandin balance
 - **Foam cells** (placental atherosclerosis)
 - Microvascular coagulation → **thrombocytopenia**
 - ↑ capillary permeability → **edema and proteinuria**
- Intact endothelium is anticoagulant and blunts smooth-muscle responses (NO); injured endothelium promotes **coagulation and vasopressor sensitivity**.

Vasospasm and hypertension

- Endothelial activation → **vasospasm** → ↑ resistance → hypertension.
- Leak into interstitium; **platelets and fibrinogen deposited subendothelially**; junctional proteins disrupted. Veins are involved too.
- Maldistribution + leak → tissue **ischemia, necrosis, hemorrhage**.
- Clinical correlate: **markedly reduced blood volume** in severe preeclampsia.

Increased pressor responses

- Normal pregnancy = **refractoriness to infused vasopressors**. Early preeclampsia = **enhanced reactivity to norepinephrine and angiotensin II**; ↑ angiotensin II sensitivity **precedes** gestational hypertension.
- Paradox: women with preterm preeclampsia have **lower circulating angiotensin II**.

Mediator	Change in preeclampsia	Effect
Prostacyclin (PGI₂) (endothelial)	↓ (via phospholipase A ₂)	Loss of vasodilation
Thromboxane A₂ (platelets)	↑	↓ PGI ₂ :TXA ₂ ratio → vasoconstriction, All sensitivity. Seen from 22 wk
Nitric oxide (from L-arginine)	↓ eNOS expression → ↓ activity	NO keeps fetoplacental bed dilated; mediates PIGF/VEGF effects
Endothelin-1 (21-amino-acid vasoconstrictor)	↑ (already ↑ in normal pregnancy)	May mediate renal injury. MgSO₄ lowers ET-1 ; sildenafil lowers it in animals
Autoantibodies	Functional antibodies to endothelin and angiotensin II receptors	—

Angiogenic and antiangiogenic factors

- Placental vasculogenesis by **21 days** after conception. VEGF and angiopoietin families are most studied.
- **Angiogenic imbalance** = excess antiangiogenic factors, driven by worsening uteroplacental **hypoxia**.
- **sFlt-1** = soluble VEGF receptor variant → binds and **lowers free PIGF and VEGF** → endothelial dysfunction.
 - Rises **months before** clinical disease, earlier with early-onset disease; high 2nd-trimester levels → much higher risk. Also seen in **FGR**.
- **sEng (soluble endoglin)** → blocks **TGF-β** binding to endothelial receptors (endoglin) → ↓ NO-dependent vasodilation. Also rises months before disease.
- **Both ↑ together → more severe disease**. Doubling of sFlt-1 / sEng raises risk by **39% / 74%**.
- ↑ sFlt-1 and ↓ PIGF in the 3rd trimester correlate with preeclampsia after **25 wk**.
- Levels are **not raised in fetal blood or amniotic fluid** and **dissipate after delivery**. Therapeutic **apheresis** of sFlt-1 is preliminary.

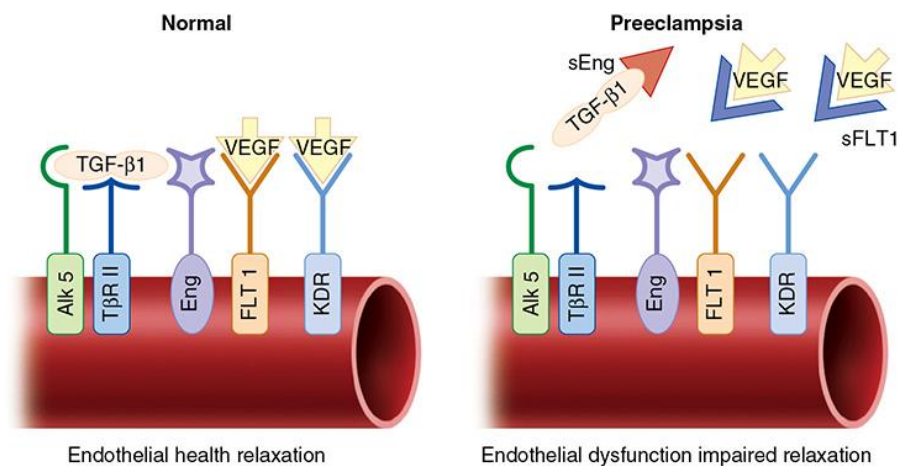


Figure 40-4. sFlt-1 mops up VEGF (and PIGF) and sEng mops up TGF-β1, so endothelial receptors (FLT1, KDR, endoglin) go unstimulated → impaired relaxation.

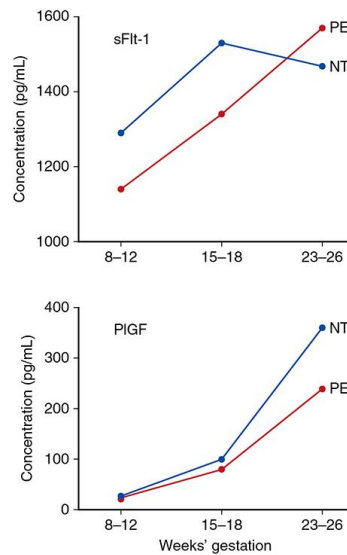


Figure 40-5. sFlt-1 rises and PlGF lags in women who develop preeclampsia; the curves diverge significantly by 23–26 wk.

4. Pathophysiology

Cardiovascular system

- Three drivers:
 1. ↑ **afterload** from hypertension
 2. **Preload**: ↓ by pathologically curtailed volume expansion, ↑ by IV crystalloid or oncotic fluids
 3. **Endothelial leak** of intravascular fluid into the extracellular space
- With clinical onset, **cardiac output falls**, partly from ↑ peripheral resistance. Some changes precede hypertension.

Myocardial function

- **Diastolic dysfunction in up to 45%** (ventricles relax and fill poorly). May persist **up to 4 yr** postpartum.
- Cause: maladaptive **ventricular remodeling** to the ↑ afterload; antiangiogenic proteins may contribute.
- Usually inconsequential, but with preexisting dysfunction (e.g., **concentric LVH from chronic hypertension**) → **cardiogenic pulmonary edema**.

Ventricular function

- Clinical cardiac function is **usually appropriate**, often **slightly hyperdynamic**. Troponin may be slightly ↑; **NT-pro-BNP** ↑ in severe disease.
- ⚠ **Aggressive IV hydration** → overtly hyperdynamic function, ↑ **PCWP**, and **pulmonary edema despite normal ventricular function**. Worsened by alveolar endothelial-epithelial leak and **low albumin** (↓ **oncotic pressure**).

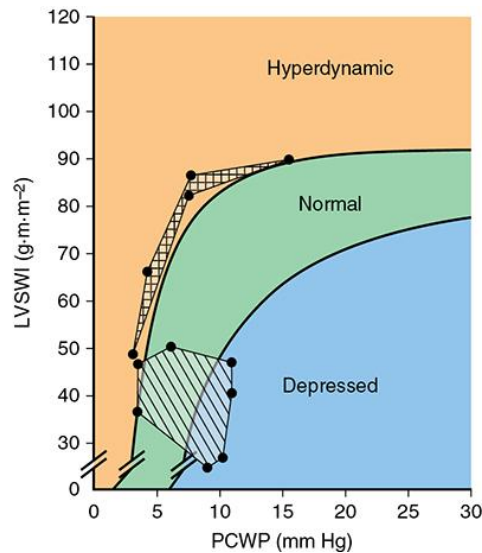


Figure 40-6. Braunwald curve: eclamptic women (boxed area) sit at the hyperdynamic edge of normal, with low PCWP; normal pregnancy (striped area) sits in the normal zone.

Blood volume

- **Hemoconcentration is a hallmark of eclampsia.**
- Nonpregnant average **3000 mL** → normal late pregnancy **4500 mL**. In eclampsia, **much or all of the expected 1500 mL excess is lost** (vasospasm + plasma leak). Less marked in preeclampsia, depending on severity.
- ⚠ Such women are **unduly sensitive to "normal" delivery blood loss**.
- After delivery, vasoconstriction reverses and volume re-expands → hematocrit falls by dilution. But **delivery blood loss** is often a substantive cause of that fall.

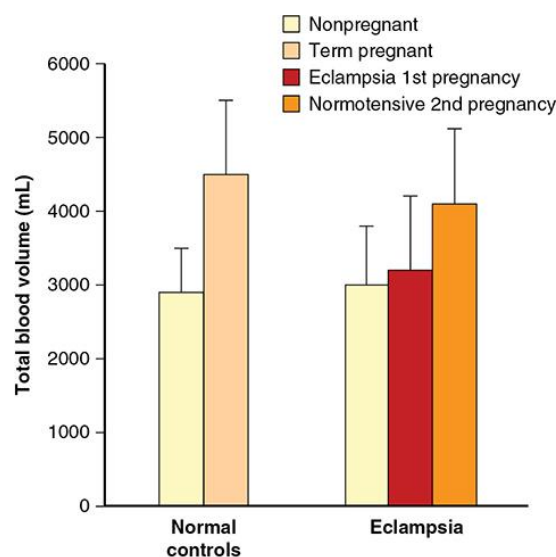


Figure 40-7. Blood volume barely rises in eclampsia (~3000 → ~3200 mL) versus ~4500 mL at term in normal pregnancy and ~4100 mL in the same women's next, normotensive pregnancy.

Hematological changes

Thrombocytopenia

- Check platelets in **any** gestational hypertension. Severity and duration of preeclampsia determine it.
- **<100,000/ μ L = severe disease**; the lower the count, the higher the maternal and fetal morbidity and mortality. Usually worsens → **delivery advisable**.
- After delivery: may fall for **~1 day**, then normalizes within **3–5 days**.
 - With HELLP it may keep falling; a nadir delayed to **48–72 h** can be wrongly attributed to a **thrombotic microangiopathy**.

- Platelet activation and $\uparrow$ α -degranulation (β -thromboglobulin, factor 4 release) $\rightarrow$ faster clearance; $\uparrow$ **platelet volume** (young platelets).
- Paradoxically $\downarrow$ **in-vitro aggregation** ("exhaustion" after in-vivo activation). $\uparrow$ platelet-bound immunoglobulins.
- **⚠ The fetus does not become thrombocytopenic** $\rightarrow$ maternal thrombocytopenia is **not a fetal indication for cesarean**.

Hemolysis

- Frequent in severe preeclampsia: $\uparrow$ **LDH**, $\downarrow$ **haptoglobin**; schizocytes, spherocytes, reticulocytosis; $\uparrow$ **RDW**.
- Mechanism: **microangiopathic hemolysis** (endothelial disruption, platelet adherence, fibrin deposition) $\pm$ serum lipid changes ($\downarrow$ long-chain fatty acids in RBCs).
- Hemolysis + thrombocytopenia + $\uparrow$ transaminases (hepatocellular necrosis) = **HELLP** (Weinstein, 1982).

Coagulation

- Subtle intravascular coagulation: $\uparrow$ factor VIII consumption, $\uparrow$ fibrinopeptides A and B and **D-dimer**, $\downarrow$ **antithrombin III, proteins C and S**.
- Usually mild and seldom significant. **Fibrinogen is not remarkably different** unless **abruption** coexists. TEG worsens as disease worsens.
- **Routine PT, aPTT and fibrinogen are NOT required** in hypertensive disorders.

Endocrine changes

- Renin, angiotensin II, aldosterone, DOC and **ANP** rise in normal pregnancy; **ANP (and pro-ANP) rise further in preeclampsia**.
- **Vasopressin**: levels similar in nonpregnant, pregnant and preeclamptic women (clearance $\uparrow$ in the latter two).

Fluid and electrolytes

- **Extracellular fluid (edema) much greater** in severe preeclampsia: endothelial injury + $\downarrow$ **plasma oncotic pressure**.
- Electrolytes are **not appreciably different**.
- After an eclamptic convulsion: $\downarrow$ **pH and HCO_3^-** from **lactic (metabolic) acidosis** plus respiratory acidosis; severity depends on lactate produced and CO_2 exhaled.

Kidney

- Renal perfusion and **GFR slightly $\downarrow$** (reversible), mostly from **afferent arteriolar resistance up to 5 $\times$** .
- **Glomerular endotheliosis** blocks filtration $\rightarrow$ **creatinine rises to nonpregnant values (~ 1 mg/dL)** or higher.
- **Uric acid $\uparrow$** beyond what the GFR explains ($\uparrow$ tubular reabsorption). **Urinary calcium $\downarrow$** .

The text prints creatinine "1 mg/mL"; the intended unit is mg/dL.

Proteinuria

Test	Abnormal threshold
24-h urine	≥ 300 mg/24 h (consensus)
12-h urine	165 mg (equivalent efficacy)
Spot urine protein:creatinine ratio	≥ 0.3
Dipstick (random)	30 mg/dL (1+) persistent

- **Nephrotic-range or worsening proteinuria is NOT a sign of severe disease**.
- P:C ratio **<0.13 – 0.15** (130–150 mg/g) $\rightarrow$ proteinuria >300 mg/d unlikely.
- P:C **<0.08 NPV 86%; >1.19 PPV 96%**. **Midrange (~ 0.3) ratios are poor** $\rightarrow$ repeat, and consider a 24-h collection if persistent.
- **Dipstick** depends on urine concentration: concentrated urine may read 1+ to 2+ with <300 mg/d. Notorious false positives and negatives.

- ⚠ Proteinuria can appear **late**: absent in **10–15% of HELLP** at presentation and in **17% of eclampsia** at the time of seizures.

Anatomy (glomerular capillary endotheliosis)

- Glomeruli **~20% enlarged**, "**bloodless**"; capillary loops dilated or contracted.
- Swollen endothelial cells** block or partly block capillary lumens; homogeneous **subendothelial protein and fibrin-like deposits**.
- Cause: **angiogenic protein "withdrawal"** (VEGF/PlGF bound by sFlt-1) → podocyte dysfunction and endothelial swelling. **Podocyturia** ↑ in eclampsia.

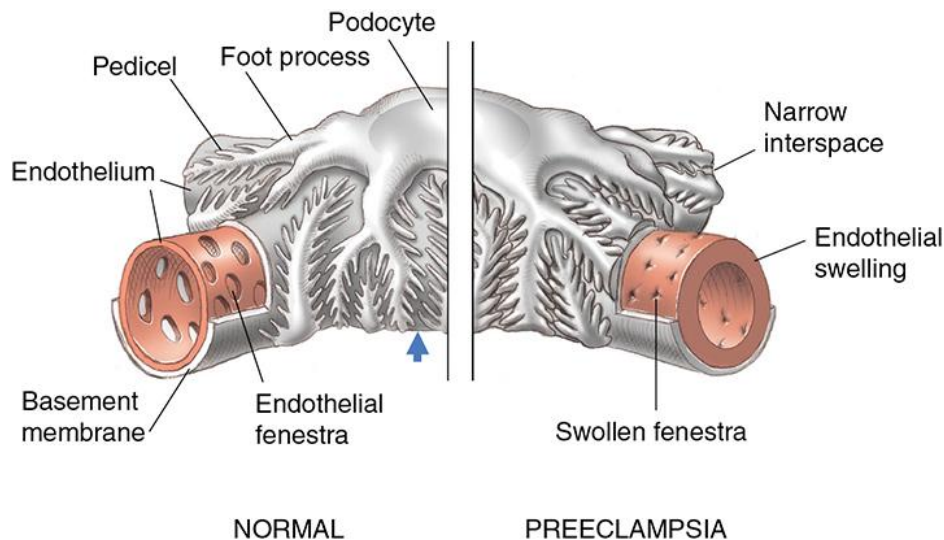


Figure 40-8. Glomerular endotheliosis: swollen endothelium with narrowed fenestrae, and podocyte pedicels that abut one another.

Acute kidney injury

Group	AKI incidence
Preeclampsia syndrome	5%
HELLP syndrome	14% (5% in another series of 183)
Severe preeclampsia (Parkland)	15%, mostly stage 1

- In preeclampsia with renal failure: **½ had HELLP, ½ had abruption**. In HELLP with AKI: half had abruption, most had PPH.
- Renal values begin to normalize **≥10 days** after delivery.
- Clinically apparent ATN is almost always from comorbid hemorrhage** → hypovolemia and hypotension.
- Urine electrolytes: preeclampsia = **intrarenal** pattern (**urine Na ↑**). Prerenal = ↑ urine osmolality, ↑ urine:plasma creatinine, **low FENa**.
- ⚠ **Oliguria is not treated with aggressive IV fluid** (risk of **pulmonary edema**) unless there is hemorrhage or loss from vomiting or fever. Crystalloid only improves output temporarily.
- Parkland switch from Ringer lactate to **normal saline** did not significantly impair renal function.

Liver

- Transaminitis** = hepatocellular injury and a **marker of severe preeclampsia**. Usually **<500 U/L** (reports **>2000 U/L**). Levels **inversely follow platelets**; both normalize within **~3 days** postpartum.
- Classic lesion: **periportal hemorrhage in the liver periphery**; some infarction with hemorrhage in **almost half** of eclamptic deaths.

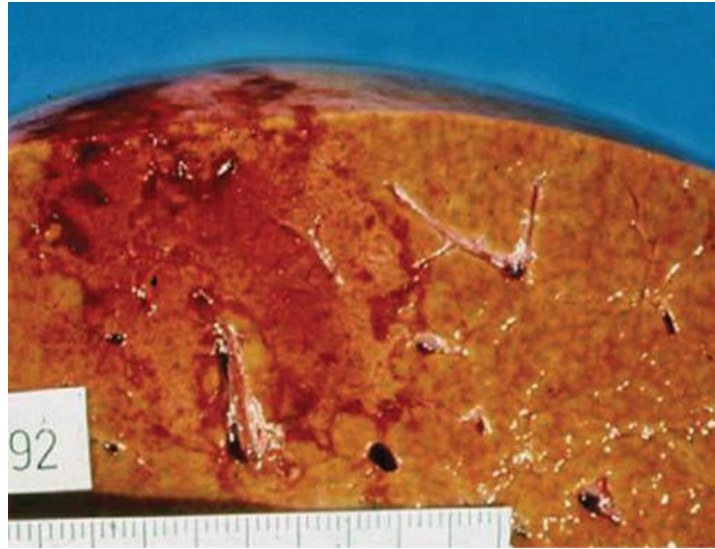


Figure 40-9. Liver in fatal HELLP: areas of ischemia, infarction and periportal hemorrhage.

Clinical manifestations

1. **Pain** (moderate–severe RUQ or midepigastria pain and tenderness) = **severe disease**, usually with $\uparrow$ AST/ALT.
2. Infarction may be extensive yet clinically silent (MRI: acute liver injury in **14 of 16** HELLP cases; volume correlated with AST). Frank infarction is unusual, worsened by **hemorrhage-related hypotension**; occasionally **"shock liver"** (hepatic failure).
3. **Hepatic hematoma** $\rightarrow$ **subcapsular hematoma** $\rightarrow$ may **rupture**.

Hepatic hematoma

- Diagnose with **CT or MRI**. Unruptured hematomas are more common than suspected, especially with HELLP.
- Management: usually **observation unless bleeding continues**; surgery or **angiographic embolization** may be lifesaving; rarely **liver transplant**.
- Review of 180 cases: **94% had HELLP**, **90% ruptured**; **maternal mortality 22%**, **perinatal mortality 31%**.



Figure 40-10. CT in severe HELLP: large subcapsular hematoma (asterisk) confluent with intrahepatic infarction/hematoma (arrowhead).

Mimics

- **Acute fatty liver of pregnancy**: also late pregnancy with hypertension, $\uparrow$ transaminases and creatinine, thrombocytopenia. **Hallmark = marked liver dysfunction**; overall liver function is **usually normal in HELLP**.
- **Pancreatitis**: not convincingly linked (**1%** in severe preeclampsia); lipase and amylase seldom $\uparrow$.

HELLP syndrome

- **H**emolysis, **E**levated **L**iver enzymes, **L**ow **P**latelets. **No universally accepted definition**, so incidence varies. Worse outcomes than preeclampsia without HELLP; some postulate a **distinct pathogenesis**.

Outcome (183 women)	Rate
Any adverse outcome	40% (2 maternal deaths)
Eclampsia	6%
Placental abruption	10%
AKI	5%
Pulmonary edema	10%
Others	Stroke, hepatic hematoma, coagulopathy, ARDS, sepsis

- Concurrent eclampsia in **10%** (693 women).

HELLP vs preeclampsia (Sep, 2009)	HELLP	Preeclampsia
Eclampsia	15%	4%
Preterm birth	93%	78%
Perinatal mortality	9%	4%

Central nervous system

Neuroanatomical lesions (autopsy)

- Brain pathology caused only **~1/3 of fatal cases**; **most deaths were from pulmonary edema**.
- Gross intracerebral hemorrhage in **up to 60%** of eclamptic women, fatal in only **half** of these.
- Principal lesions: **cortical and subcortical petechial hemorrhages**. Classic microscopic lesion: **fibrinoid necrosis of the arterial wall** with perivascular microinfarcts and hemorrhages.
- Also nonhemorrhagic "softening," white-matter hemorrhage, subcortical edema, and **basal ganglia or pontine hemorrhage** that may rupture into the ventricles.



Figure 40-11. Autopsy brain slice: fatal hypertensive hemorrhage rupturing into the ventricles in an eclamptic primigravida.

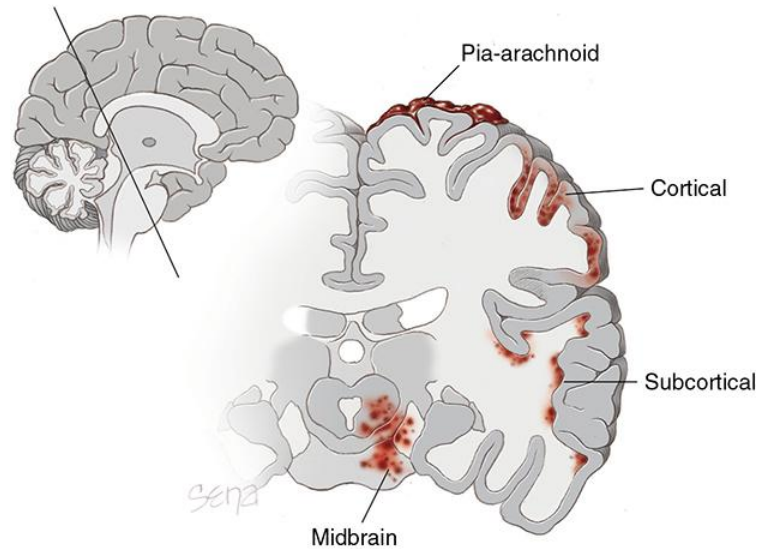


Figure 40-12. Sites of hemorrhage and petechiae in eclampsia: pia-arachnoid, cortical, subcortical and midbrain.

Cerebrovascular pathophysiology: two theories (endothelial dysfunction in both)

Theory	Mechanism	Evidence
Overregulation	Acute severe hypertension → cerebral vasospasm → infarction	Little objective evidence
Forced dilation (loss of autoregulation)	Sudden BP rise exceeds autoregulation → forced vasodilation/constriction in arterial boundary zones → ↑ capillary hydrostatic pressure, hyperperfusion, leak through tight junctions → vasogenic edema	Supported

- Most likely a **combination**: an endothelial leak at BP far **lower** than usually causes vasogenic edema, plus loss of upper-limit autoregulation.
- Result: **PRES (posterior reversible encephalopathy syndrome)**, mainly **occipital and parietal cortex**.

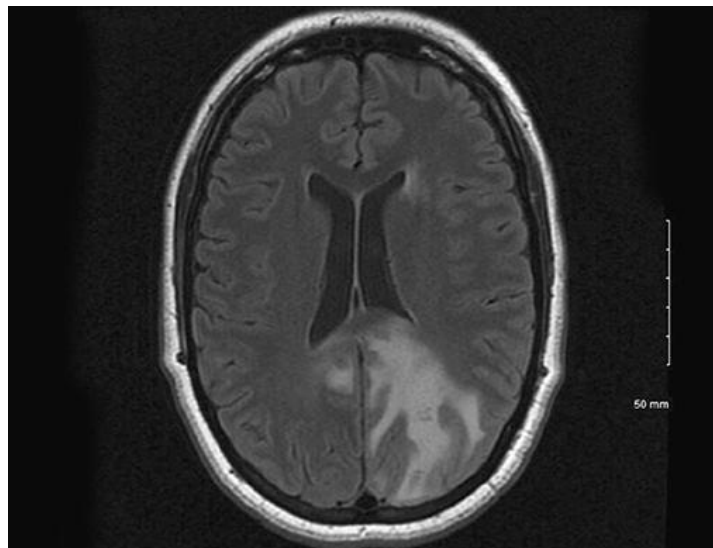


Figure 40-13. T2-FLAIR MRI in eclampsia: occipital hyperintensity typical of PRES.

Cerebral blood flow

- Nonpregnant autoregulation protects against hyperperfusion up to a **MAP of 160 mm Hg**, far above the pressures in most eclamptic women → autoregulation must be altered in preeclampsia (impaired autoregulation shown; normal again **2–3 yr** postpartum).
- Normal pregnancy: CBF unchanged in the first two trimesters, then ↓ **20% in the 3rd trimester**. **Severe preeclampsia: CBF higher** than in normotensive women.
- Synthesis: eclampsia = **cerebral hyperperfusion** forcing fluid into the interstitium through damaged endothelium → **perivascular edema**.

Neurological manifestations (each = severe disease)

- **Headache and scotomata** from occipital hyperperfusion. Before eclampsia: **headache in up to 75%, visual changes in 20–30%**.
 - The headache typically **does not respond to usual analgesics** but often **improves with magnesium sulfate**.
- **Convulsions are diagnostic of eclampsia**. Prolonged seizures can cause lasting brain injury.
- **Cognitive decline** 5–10 yr after eclampsia.

Neuroimaging

- **CT**: hypodense lesions at the **gray-white junction**, mainly **parietooccipital** (also frontal, inferior temporal, basal ganglia, thalamus) = petechial hemorrhage and edema. Diffuse edema → compressed ventricles, risk of **transtentorial herniation**.
- **MRI: T2 hyperintense** cortical/subcortical parietooccipital lesions = PRES (also basal ganglia, brainstem, cerebellum).
 - PRES is **almost universal in eclampsia** and in **~20% of severe preeclampsia**.
 - **~1/4** of eclamptic lesions show **restricted diffusion = infarction** with persistent findings.

Visual changes and blindness

- ↓ retinal arteriolar and venular caliber + visual cortex involvement → scotomata, blurring, diplopia; improve with **MgSO₄ and/or BP lowering**.
- **Blindness**: rare with preeclampsia alone but in **up to 15% of eclampsia**; may start **≥1 week after delivery**; usually reversible.

Site	Features
Occipital visual cortex (amaurosis)	Extensive occipital vasogenic edema on MRI. Parkland: lasted 4 h to 8 days , resolved completely in all 15
Lateral geniculate nuclei	—
Retina	Ischemia, infarction (Purtscher retinopathy , rare) or serous detachment (usually unilateral, seldom total loss; asymptomatic detachment relatively common). Retinal artery occlusion → may be permanent



Figure 40-14. Purtscher retinopathy: scattered yellowish opaque retinal lesions (arrows) from choroidal ischemia and infarction.

Cerebral edema

- Parkland: **10 of 175** eclamptic women had symptomatic cerebral edema. Symptoms from lethargy, confusion and blurred vision to obtundation and coma, often waxing and waning.
- **3 became comatose** with transtentorial herniation; **1 died**. Mental status correlated with imaging.
- ⚠️ Very sensitive to sudden BP surges → **careful BP control is essential**.

Uteroplacental perfusion

- Poor uteroplacental perfusion is a major cause of **perinatal morbidity and mortality**. Direct measurement is hard; **uterine artery Doppler** (systolic:diastolic waveforms) is the surrogate.
- Normally impedance falls markedly once placentation is complete; with abnormal placentation **high resistance persists**.
- **Uterine artery "notching"** → ↑ risk of preeclampsia or FGR, but **low predictive value except for early-onset severe disease** (MFMU).
- Only $\frac{1}{3}$ of women with severe 3rd-trimester disease, and $\frac{1}{3}$ with **HELLP**, have abnormal uterine artery velocimetry.
- Abnormal waveforms correlate with **fetal** involvement → useful to predict **FGR, not preeclampsia**. No other waveform is clinically suitable.

Fetal-growth restriction

- A severity indicator; **usually confined to fetuses of women destined for severe preeclampsia**.
- FGR pregnancies share preeclampsia-like hemodynamics: ↑ MAP, ↑ SVR, ↓ cardiac output, ↑ uterine artery PI. Fetuses of preeclamptic mothers show **cardiac remodeling** like FGR fetuses.

5. Prediction

- Overall: **poor sensitivity and PPV. No screening test is reliable, valid and economical**. Multivariable algorithms (1st-trimester sFlt-1, midpregnancy) are **not verified for widespread use**.

Test category	Examples	Performance
Pressor tests	Roll-over test (28–32 wk: left lateral → supine, BP rise = positive), isometric handgrip , angiotensin II infusion	Sensitivity 55–70% , specificity ~85% ; cumbersome
Uterine artery Doppler	Resistance, notching	Poor for preeclampsia (better for FGR)
Fetal-placental endocrine analytes	Many serum analytes	None clinically beneficial
Serum uric acid	↓ clearance (↓ GFR, ↑ reabsorption, ↓ secretion)	Sensitivity 0–55% , specificity 77–95% ; seldom used
Microalbuminuria	Isolated gestational proteinuria is a risk factor	Sensitivity 7–90% , specificity 29–97%
Endothelial / oxidative	Fibronectin (not useful), platelet volume (early predictor), coagulation markers (overlap), lipid peroxides, iron/ferritin, resistin, homocysteine, lipids, vitamins C/E	None sufficiently predictive
Angiogenic imbalance	↓ VEGF , PlGF ; ↑ sFlt-1 , sEng (and ratios) before disease	Best, especially for early-onset disease
Others	cfDNA (no correlation, MFMU), HbA1c, cystatin C, 1st-trimester placental volume, proteomics/metabolomics	Preliminary

- **Angiogenic markers as diagnostic adjuncts**: help separate preeclampsia from **mimics** (chronic hypertension, CKD, SLE, immune thrombocytopenia) and **mild from severe** disease; likely future role in 1st-trimester screening.

6. Prevention

- **With the possible exception of aspirin, nothing is convincingly and reproducibly effective.**

Table 40-4. Some Methods to Prevent Preeclampsia That Have Been Evaluated in Randomized Trials

Category	Methods
Dietary manipulation	Low-salt diet, calcium or fish oil supplementation
Exercise	Physical activity, stretching
Cardiovascular drugs	Diuretics, antihypertensive drugs
Antioxidants	Ascorbic acid (vitamin C), α -tocopherol (vitamin E), vitamin D
Antithrombotic drugs	Low-dose aspirin, aspirin/dipyridamole, aspirin + heparin, aspirin + ketanserin

Modified from Jim, 2017; Staff, 2015.

Diet and lifestyle

Intervention	Evidence
Low-salt diet	Ineffective (RCT)
Exercise	Linked to lower risk; few RCTs
Bed rest	Hospital bed rest RR 0.27 (cohort); home rest 4–6 h/d helped in 2 small RCTs. SMFM does not recommend reduced activity
Calcium	No benefit in >4500 low-risk nulliparas; benefit only if calcium deficient
Iodine	No association with preeclampsia (subclinical hypothyroidism is a risk factor)
Folic acid 4 mg/d	No effect (~14% in both arms, ~2500 high-risk women)
Fish oil	No benefit in RCTs (seafood diet protective in one cohort)

Drugs

Agent	Evidence
Diuretics	↓ edema and hypertension but not preeclampsia (9 RCTs, >7000)
Antihypertensives in chronic hypertension	No proven reduction in superimposed preeclampsia
Vitamins C and E	No benefit (CAPPS, ~10,000 low-risk nulliparas, and others)
Statins (pravastatin)	↑ heme oxygenase-1 → inhibits sFlt-1 release; animal data; pilot done, MFMU RCT planned
Metformin	Inhibits HIF-1 α → ↓ sFlt-1 and sEng; ↓ severe preeclampsia in one preliminary prediabetic study
LMWH	No reduction in recurrent preeclampsia, abruption or FGR (IPD meta-analysis, 963 women); may help high-risk women $\pm$ aspirin
Aspirin + enoxaparin vs aspirin (prior early-onset)	Similar outcomes
Aspirin + heparin	Mitigates thrombosis with lupus anticoagulant

Low-dose aspirin

- **50–150 mg/d** inhibits platelet **thromboxane A₂** with minimal effect on vascular prostacyclin.
- Evidence:
 - MFMU (1998): no significant benefit.
 - **ASPRE (Rolnik, 2017)**: >1600 high-risk women, **11–14 wk → 36 wk**: preterm preeclampsia **1.6% vs 4.3%**.
 - Started **<16 wk**: **~60%** ↓ preeclampsia and FGR (Roberge, 2017); ↓ **preterm but not term** preeclampsia (Roberge, 2018).

◦ IPD meta-analysis: only **~10%** reduction, whether started before or after 16 wk (Meher, 2017).

• **USPSTF** recommends low-dose aspirin for high-risk women. **ACOG: start between 12 and 28 wk.**

High risk: ≥ 1 → give aspirin	Moderate risk: > 1 → consider aspirin
Prior preeclampsia	Nulliparity
Chronic hypertension	Age > 35 yr
Overt diabetes	Obesity
Renal disease	Family history of preeclampsia
Autoimmune disorders	Vulnerable sociodemographics
Multifetal gestation	Prior low-birthweight or growth-restricted neonate

Pearl: In practice (ACOG/USPSTF), the aspirin dose used in the US is **81 mg daily**, started ideally before 16 wk and continued until delivery.

Quick-Recall: Organ Involvement in Preeclampsia

System	Key lesion / mechanism	Clinical marker	Must-know number
Placenta	Shallow trophoblast invasion; atherosis	FGR, abruption, abnormal uterine artery Doppler	Myometrial arterioles $\frac{1}{2}$ normal diameter
Endothelium	$\uparrow$ sFit-1 , $\uparrow$ sEng , $\downarrow$ PIGF/VEGF ; $\downarrow$ PGI_2 , $\uparrow$ TXA_2 , $\uparrow$ ET-1 , $\downarrow$ NO	Hypertension, edema, proteinuria	Diverge by 23–26 wk
Heart	Diastolic dysfunction; hyperdynamic function	Pulmonary edema with fluids	Diastolic dysfunction $\leq 45\%$
Blood volume	Hemoconcentration	Sensitive to delivery blood loss	Expected +1500 mL lost
Platelets	Consumption, activation	$< 100,000$ = severe	Normal by 3–5 d postpartum
RBCs	Microangiopathic hemolysis	$\uparrow$ LDH , $\downarrow$ haptoglobin , schistocytes	—
Coagulation	Mild DIC-like changes	Fibrinogen normal unless abruption	Routine PT/aPTT not needed
Kidney	Glomerular endotheliosis ; $\uparrow$ afferent resistance	Proteinuria, $\uparrow$ creatinine, $\uparrow$ urate, $\downarrow$ urinary Ca	Creatinine > 1.1 or doubled
Liver	Periportal hemorrhage , infarction, subcapsular hematoma	RUQ pain, transaminases $2\times$ normal	Rupture: 22% maternal mortality
Brain	Loss of autoregulation → vasogenic edema (PRES)	Headache, scotomata, seizures	PRES in ~20% of severe preeclampsia
Eye	Occipital cortex edema; retinal detachment/infarction	Blindness	$\leq 15\%$ of eclampsia; usually reversible

Key Numbers to Memorize

- Hypertensive disorders **$\leq 10\%$** of pregnancies; preeclampsia **5–8%** (nulliparas **3–10%**, multiparas **2–5%**); **7%** of US pregnancy-related deaths
- Hypertension **$\geq 140/90$** (Korotkoff V); severe **$\geq 160/110$** ; chronic = before pregnancy or **< 20 wk**; gestational hypertension resolves by **12 wk** postpartum
- Proteinuria **≥ 300 mg/24 h**, P:C **≥ 0.3** , dipstick **1+** (30 mg/dL); 12-h **165 mg**
- Platelets **$< 100,000$** ; creatinine **> 1.1 mg/dL** or doubled; transaminases **$2\times$ normal**
- Early onset **< 34 wk**; preterm **< 37 wk**

- $\sim 1/2$ of gestational hypertension becomes preeclampsia; **10%** of eclampsia before proteinuria; **$\sim 10\%$** of seizures **>48 h** postpartum
- Superimposed preeclampsia in **20–50%** of chronic hypertension (after **24 wk**)
- Highest RR: **prior preeclampsia 8.4**, **CHTN 5.1**, diabetes 3.7; trisomy 13 $\rightarrow$ **30–40%**
- Eclampsia **1 in 2000–3000** deliveries
- Blood volume **3000 $\rightarrow$ 4500 mL** normally; eclampsia loses the **1500 mL** excess
- AKI: **5%** preeclampsia, **14%** HELLP, **15%** severe preeclampsia (Parkland)
- Hepatic hematoma: **94%** HELLP, **90%** ruptured, maternal death **22%**, perinatal death **31%**
- Pre-eclamptic headache **75%**, visual symptoms **20–30%**; blindness **$\leq 15\%$** of eclampsia
- Aspirin **50–150 mg/d**, start **12–28 wk** (ideally **<16 wk**); ASPRE **1.6% vs 4.3%**